



Bahir Dar University

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Proposal submission

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Title 1: Tessessir (ፕሰሰር): An AI-Powered Intelligent Matching System for Diaspora-Led Volunteer Services

Introduction:

Access to skilled mentorship, professional training, and specialized support is limited for students, disabled individuals, and underserved communities in Ethiopia, while many Ethiopian professionals abroad are willing to contribute their expertise and resources. Currently, volunteer efforts are fragmented, informal, and difficult to coordinate, resulting in underutilized skills and missed opportunities. Tessessir proposes an AI-powered platform that connects diaspora professionals with verified local beneficiaries, facilitating structured volunteer programs, mentorship, training, counseling, and donations, while ensuring transparent tracking, intelligent matching, and measurable social impact.

Statement of the Problem

Communities in Ethiopia face challenges in accessing professional guidance and volunteer support due to:

- Lack of a centralized platform connecting diaspora volunteers with local beneficiaries.
- Difficulty in finding verified mentors for students and institutions.
- Fragmented and non-transparent donation and volunteer systems, reducing trust and efficiency.

Objectives

The goal of Tessessir is to create a centralized, intelligent, and transparent volunteer coordination system:

- Volunteer and Beneficiary Registration: Develop a secure system for diaspora professionals and local beneficiaries to register and verify their identities and credentials.
- AI Matching Engine: Implement AI algorithms to intelligently match volunteers with beneficiaries based on skills, needs, availability, and priority.
- Donation and Resource Tracking: Integrate a transparent system to track financial and material contributions.

Significance of the Project

- Enable efficient, transparent, and impactful diaspora-led volunteer programs.
- Provide students and underserved communities access to global expertise and mentorship.
- Foster trust through transparent donation tracking and accountability.
- Serve as a scalable model for digital volunteer coordination in Ethiopia, enhancing national development and social empowerment.

Title 2: “Gibi-Hub” - Distributed P2P Educational Repository

Introduction

Access to large educational resources is a major challenge for students in public universities in Ethiopia. Slow or unreliable internet makes downloading video lectures, high-resolution textbooks, and datasets difficult. Gibi-Hub proposes a peer-to-peer (P2P) content delivery system that works over a university’s local network, allowing students to efficiently share and download educational resources without heavy reliance on internet connectivity.

Statement of the Problem

Students face barriers to effective learning due to internet limitations, including:

- Slow and unreliable internet, which makes accessing large educational files impractical.
- Dependence on cloud-based resources, which may not be accessible at all times.
- Lack of offline content distribution systems, causing students to miss out on critical learning materials.

Objectives

The goal of Gibi-Hub is to create a seamless educational content network:

- **Server Development:** Build a centralized server that maintains a database of available educational materials within the university LAN.
- **P2P Client:** Develop a BitTorrent-like client enabling students to download and share files efficiently across the local network.
- **Offline-First Caching:** Implement a system to automatically update content when internet bandwidth is sufficient, ensuring resources are available at all times.
- **Demonstration:** Showcase high-speed local downloads (e.g., 100MB/s) without relying on external internet, validating the effectiveness of the system.

Significance of the Project

- Provide students with consistent, high-speed access to critical educational resources.
- Reduce dependence on unstable or slow internet connections.
- Foster collaborative sharing of learning materials within academic institutions.
- Serve as a model for other universities and local networks across Ethiopia, potentially improving educational outcomes nationwide.

Title 3: Construction Material Price & Supplier Marketplace

Introduction

The construction sector in Ethiopia is growing rapidly, but contractors face challenges with fluctuating material prices and inefficient procurement processes. Daily changes in the prices of cement, steel, and other materials lead to wasted time and resources. The proposed solution is a real-time construction materials marketplace that consolidates supplier information, updates prices daily, provides bulk order options, and offers analytical tools to optimize procurement decisions.

Statement of the Problem

Contractors and construction companies face operational challenges including:

- Rapidly changing material prices, creating uncertainty and complicating project planning.
- Time-consuming supplier comparisons, which reduce productivity.
- Lack of transparent pricing systems, leading to inefficient procurement and material wastage.

Objectives

The project aims to simplify and modernize construction procurement:

- Marketplace Development: Build a web/mobile platform to display real-time construction material prices.
- Supplier & Order Management: Implement features such as supplier ratings, bulk order management, and notifications for price changes.
- Analytical Tools: Provide graphs and reports showing price trends to help contractors make informed decisions.
- Efficiency Demonstration: Show how the system reduces procurement time, improves cost management, and minimizes material wastage.

Significance of the Project

This project will:

- Increase transparency and efficiency in material procurement.
- Help contractors make cost-effective decisions with real-time data.
- Reduce material wastage and mismanagement.
- Support digital transformation in Ethiopia's construction sector, making day-to-day operations faster, more reliable, and easier to manage.